

Lecture 2: Contiguous Memory Allocation

Fundamentals of Operating System Design (CEUC301)

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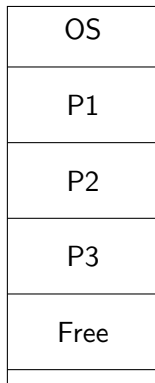
DEPSTAR, CHARUSAT

Learning Outcomes

- Explain contiguous memory allocation.
- Differentiate fixed and dynamic partitioning.
- Identify internal and external fragmentation.
- Explain compaction.

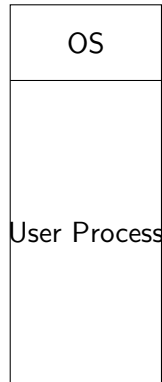
Contiguous Memory Allocation

- Each process occupies one continuous block of memory.
- Simple to implement.
- Used in early operating systems.
- Fast address translation.



Single Partition Allocation

- Memory divided into OS area and one user area.
- Only one user process executes at a time.
- Simple but poor CPU utilization.

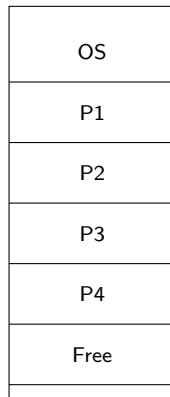


Multiple Partition Allocation

- Memory divided into multiple partitions.
- Multiple processes reside in memory simultaneously.
- Partitions may be fixed or variable.

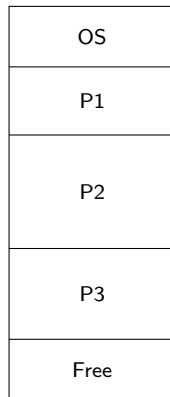
Fixed Partition Allocation

- Main memory is divided into fixed-size partitions.
- One process occupies one partition.
- Number of processes is limited by the number of partitions.
- May cause internal fragmentation.



Dynamic (Variable) Partition Allocation

- Partition size depends on process size.
- Better memory utilization.
- Number of partitions changes dynamically.
- May cause external fragmentation.



Fixed vs Dynamic Partitioning

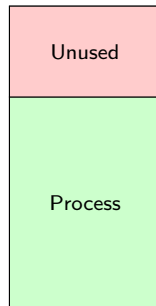
Feature	Fixed Partition	Dynamic Partition
Partition Size	Fixed	Variable
Memory Utilization	Lower	Better
Internal Fragmentation	Yes	No
External Fragmentation	No	Yes
Implementation	Simple	More Complex

Internal Fragmentation

- Unused space **inside** an allocated partition.
- Occurs mainly in fixed partition allocation.
- Process size is smaller than partition size.

Example:

- Partition = 100 KB
- Process = 70 KB
- Internal Fragmentation = 30 KB

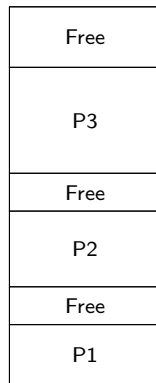


External Fragmentation

- Free memory is split into many small blocks.
- Total free memory may be sufficient.
- No single block is large enough.

Example:

- Free blocks: 20 KB, 15 KB, 25 KB
- Request: 40 KB
- Allocation fails.



Compaction

- Compaction combines scattered free spaces.
- Eliminates external fragmentation.
- Requires relocation support.
- Increases execution overhead.

Before:

$[P1][F][P2][F][P3][F]$

After:

$[P1][P2][P3][\text{Free}]$

Worked Example

Memory Size: 1000 KB

OS	100 KB
P1	200 KB
P2	250 KB
Free	450 KB

If a new process of 300 KB arrives:

- Allocation succeeds.
- Remaining free memory = 150 KB.

Summary

- Contiguous allocation stores each process in one continuous memory block.
- Single and multiple partition allocation are basic approaches.
- Fixed partitioning causes internal fragmentation.
- Dynamic partitioning may cause external fragmentation.
- Compaction reduces external fragmentation.

Questions

- 1 Explain contiguous memory allocation.
- 2 Compare fixed and dynamic partition allocation.
- 3 Differentiate internal and external fragmentation.
- 4 What is compaction? Why is it needed?
- 5 Solve a memory allocation problem using contiguous allocation.